

Understanding Data and Measurement Fundamentals for Domain Analysis

Task 7, Day 1 — Data Understanding and Measurement Basics

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Domain: Software / ML

Task Number: 7 (Week 2, Day 1)

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Submission Date: 02 July 2026

1. Introduction

This report builds the theoretical foundation required before working with pandas, feature engineering, visualization, or machine learning. It focuses on what a dataset represents, how measurements are structured, how variable types differ, and why domain context and units matter before any numerical analysis is attempted. Examples are drawn from Biochemistry, Electronics, and Mechanical domains to keep the discussion grounded rather than purely abstract.

2. Observation vs. Variable

An observation is a single recorded instance of data collection — one row in a dataset, corresponding to one measurement event, one specimen, or one time sample. A variable, by contrast, is a property being measured across all observations — it corresponds to a column. For example, if an ultrasonic sensor is tested at ten different distances, each individual test at a specific distance is one observation, while "target distance" and "sensor output voltage" are variables recorded for every observation. Confusing the two leads to structural errors, such as treating a single reading as if it represents an entire trend, or treating a variable name as if it were a data point.

3. Categorical vs. Numerical Data

Categorical data represents labels or groupings — values that describe a quality or category rather than a quantity, and cannot be meaningfully added or averaged. Numerical data represents quantities that result from counting or measurement, and support arithmetic operations such as averaging, differencing, and ratio comparison. The distinction matters because the two types require different statistical treatment: categorical data is summarised using counts, while numerical data is summarised using measures such as mean, median, and standard deviation.

4. Domain Examples of Categorical and Numerical Data

Categorical Data Examples

- Biochemistry: blood type (A, B, AB, O) — a classification label with no inherent numeric order.
- Electronics: component type on a test bench (resistor, capacitor, sensor) — a discrete label identifying the part.
- Mechanical: material grade of a test specimen (mild steel, aluminium 6061, cast iron) — a categorical description of composition.

Numerical Data Examples

- Biochemistry: enzyme concentration in mg/mL — a continuous quantity from a measurement.

- Electronics: sensor output voltage in volts, or current in milliamps — continuous measured quantities.
- Mechanical: tensile stress in MPa, or strain as a percentage — continuous measured quantities from a load test.

5. Raw Data vs. Processed Data

Raw data is the unaltered output captured directly from an instrument or logging process — it may contain noise, missing entries, duplicate rows, or physically implausible readings caused by instrument glitches. Processed data is raw data that has been cleaned, validated, and possibly transformed: missing values handled, impossible values removed or corrected, units standardised, and the data restructured into a form suitable for analysis. For instance, a calibration log straight off an ultrasonic sensor is raw data; the same log after removing a negative-voltage glitch and dropping incomplete rows is processed data. Analysis performed directly on raw data without this step risks producing statistics that are skewed by errors rather than by the underlying phenomenon.

6. Measurement Units and Why They Matter

A measurement unit defines the scale and reference against which a numerical value is interpreted — a number alone is meaningless without it. Stating a stress value as "250" is uninformative until it is known whether the unit is MPa, psi, or N/mm², since each implies a different physical magnitude. Units matter because they encode the physical meaning of a number, enable correct comparison across instruments or teams, and are essential for any downstream calculation, such as converting a sensor voltage into a physical distance or a stress value into a strain estimate using a known material constant.

7. Why Comparing Values Across Different Units Is Dangerous

Comparing numerical values without confirming that they share the same unit can silently produce incorrect conclusions, because the numbers may look comparable in magnitude while representing entirely different physical quantities. A classic failure mode is comparing a distance recorded in centimetres against one recorded in millimetres without conversion — the values differ by a factor of ten, but a naive comparison would misinterpret which reading is larger or smaller. In an engineering context, mixing voltage readings in millivolts with readings in volts, or stress values in MPa with values in psi, can lead to a design or safety decision based on a number that is off by orders of magnitude.

8. Basic Summary Statistics: Mean, Median, Minimum, Maximum, Range

The mean is the arithmetic average of all values in a dataset and represents its central tendency, but it is sensitive to extreme values. The median is the middle value when data is sorted in order, and it is more robust to outliers than the mean. The minimum and maximum are the smallest and largest observed values, marking the boundaries of the data. The range is the

difference between maximum and minimum, giving a first, coarse sense of how spread out the values are. These five statistics together form the minimum starting summary that should be computed before any deeper analysis of a numerical variable.

9. When the Mean Can Be Misleading

The mean can be misleading when a dataset contains outliers, is strongly skewed, or is multimodal (has more than one cluster of typical values). A single extreme reading — for example, an instrument glitch producing a stress value far above the rest of a tensile test — can pull the mean noticeably away from where most of the data actually sits, while the median remains largely unaffected. In such cases, reporting the mean alone without also checking the median, the spread, and a visual distribution (histogram or boxplot) risks presenting a summary that does not reflect the typical behaviour of the system being measured.

10. What Should Be Inspected Before Analyzing a Dataset

Before any formal analysis, a dataset should be inspected for: the presence and pattern of missing values; duplicate rows; the data type of each column (numerical vs categorical) and whether it matches expectation; the measurement unit of each numerical column; physically implausible values (e.g. negative voltage, negative mass); and potential outliers (via a boxplot or z-score/IQR check). This first-pass inspection, sometimes called exploratory data analysis, determines whether the dataset is trustworthy enough to support the conclusions that will later be drawn from it.

11. Questions to Ask Before Analyzing Domain Data From Another Team

- What instrument or method was used to collect this data, and what is its known noise or error range?
- What are the units of every numerical column, and are they consistent throughout the file?
- Were there any known experimental conditions (temperature, calibration state, specimen batch) that could affect the readings?
- Have missing values, duplicate rows, or known instrument glitches already been removed, or is this raw data?
- Is there a data dictionary or README explaining what each column represents?
- What is the expected range of "normal" values, so that implausible readings can be identified?

12. Domain Thinking: A Dataset Is More Than a Table

A numerical value only becomes meaningful once it is connected to its variable name, its measurement unit, the experimental condition under which it was recorded, and the domain context surrounding it. The number "0.014" is meaningless in isolation; it becomes informative

only when identified as a sensor output voltage, in volts, under stable bench conditions, as part of an ultrasonic sensor's blank calibration. This is why the first step in any analysis task — before writing a single line of pandas code — must be understanding what each column represents, not simply what values it contains.

13. Domain Examples Overview

Table 1 below summarises one categorical and one numerical example from each of the three domains referenced throughout this report, and is referenced as supporting context for Sections 4 and 12 above.

Domain	Categorical Example	Numerical Example
Biochemistry	Blood type (A, B, AB, O)	Enzyme concentration (mg/mL)
Electronics	Component type (resistor, sensor)	Sensor output voltage (V)
Mechanical	Material grade (mild steel, Al 6061)	Tensile stress (MPa)

Table 1. Categorical and numerical examples across three engineering domains.

14. Conclusion

Before any dataset can be responsibly analysed, its structure must be understood: which columns are observations versus variables, which are categorical versus numerical, what units apply, and what condition the data was collected under. Raw data must be distinguished from processed data, summary statistics must be interpreted with awareness of their limitations (particularly the mean's sensitivity to outliers), and a short checklist of questions should be run before trusting data handed over by another team. These foundations are what make later pandas-based analysis, visualization, and modelling meaningful rather than mechanical.

15. References

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